TEAM #124 · MEGATHON X · LLOYDS PROBLEM STATEMENT 2

Ibasho

See how a building will live in its city.

A studio for sun, daylight, energy, carbon, codes and retrofits, from a single desk to a whole city.

Ibasho: a place where you belong

PROBLEM

A green score for one house, cut off from its street, misses most of the story.

Neighbours

A top-floor flat in Hyderabad cooks in May. The same flat beside a taller block does not.

Use

On the same floor plate, a lab uses about 1.6 times the energy of an office.

Place

A London terrace and a Gachibowli block need different physics, codes and tariffs.

PROBLEM

Sustainability starts at a desk and grows.

Desk › Room › Floor › Building › Community › Area › City

A homeowner, an architect, a campus estates team and a business ask the same question at different scales. Ibasho answers it at every step of the ladder.

APPROACH

Published physics, run on a real city, explained in plain words.

Place
City from OpenStreetMap, hourly weather

›

Building
Storeys, rooms, walls, windows, shades

›

Calculate
Sun, radiation, daylight, energy, codes

›

Evidence
Ranges, sources and method for each number

›

Explain
Narrator and agent read, never compute

Rough answer in under a second, then refined, then final.

Unknowns stay as ranges, such as a neighbour's height or a wall's insulation.

Packages recomputed whole, never summed, so measures interact honestly.

APPROACH

Checked, not promised.

Every number has a source

Each number links to its inputs, its method and its published source. Unknowns stay as ranges, never guesses.

An independent checker

A checker written in the Lean 4 proof assistant re-derives savings, carbon, bills and ranking for every building result.

The AI only explains

Any sentence that writes a number of its own is rejected. A measure you rule out never returns.

Private by design

It runs on your own machine. No account, and no data leaves the device.

SOLUTION

A design and analysis studio for any building, in its real city.

Pick a place, pick or draw a building, set it among its real neighbours. See sun and shade, daylight, energy by end use, bills, carbon, codes and the green rating path. Then compare changes.

Eight building types · Hyderabad and London · runs on your own machine



SOLUTION

Fix what stands. Design what comes next.

Reactive

For a building that exists and needs to improve. Which retrofits suit it, in what order, what each saves in electricity and gas, and what it does to carbon and value.

The brief's retrofit question

Proactive

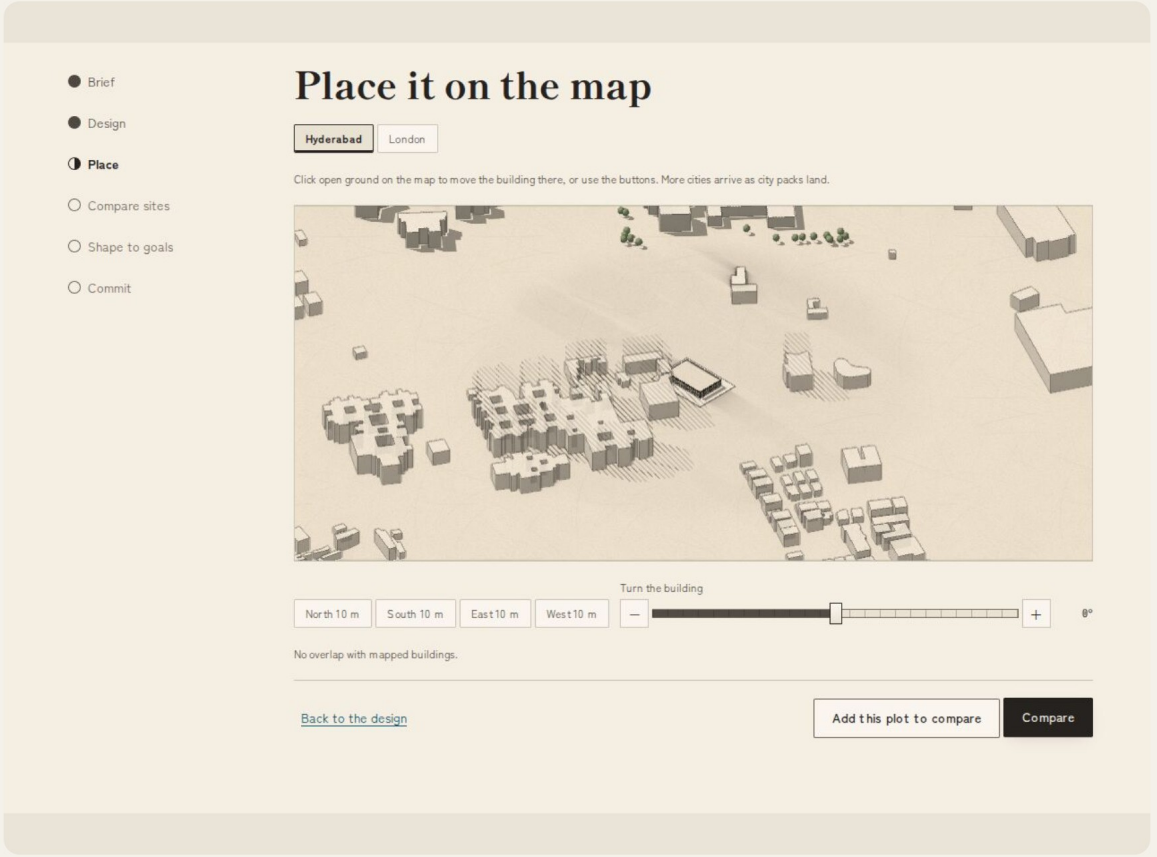
For a building still on paper, where a change costs almost nothing. Draw it, place it on a real plot, see its sun, daylight, energy and code standing before a brick is laid.

Design before you build

SOLUTION

One tool, from a homeowner to a campus.

- **Homeowners and landlords:** what to fix first, and what it saves
- **Architects:** sun, daylight and energy before anything is built
- **Estates and facility teams:** every building on a site, compared
- **Businesses:** an office or a lab, set among its real neighbours



SOLUTION

What you can do with Ibasho today.

Find anything by name
Places, buildings, floors and rooms

See it in its real city
Hyderabad and London, real neighbours

Sun and shadow
Any day, any hour, who shades whom

Daylight and radiation
Floor by floor, wall by wall

Energy, bills and carbon
Split by end use, in local tariffs

Codes and green ratings
What it meets, and what it is short of

What if, side by side
Glazing, shading, solar, compared

Design before you build
Brief, plan editor, place on a plot

Green Home Score
Ranked homes, retrofits, savings, value

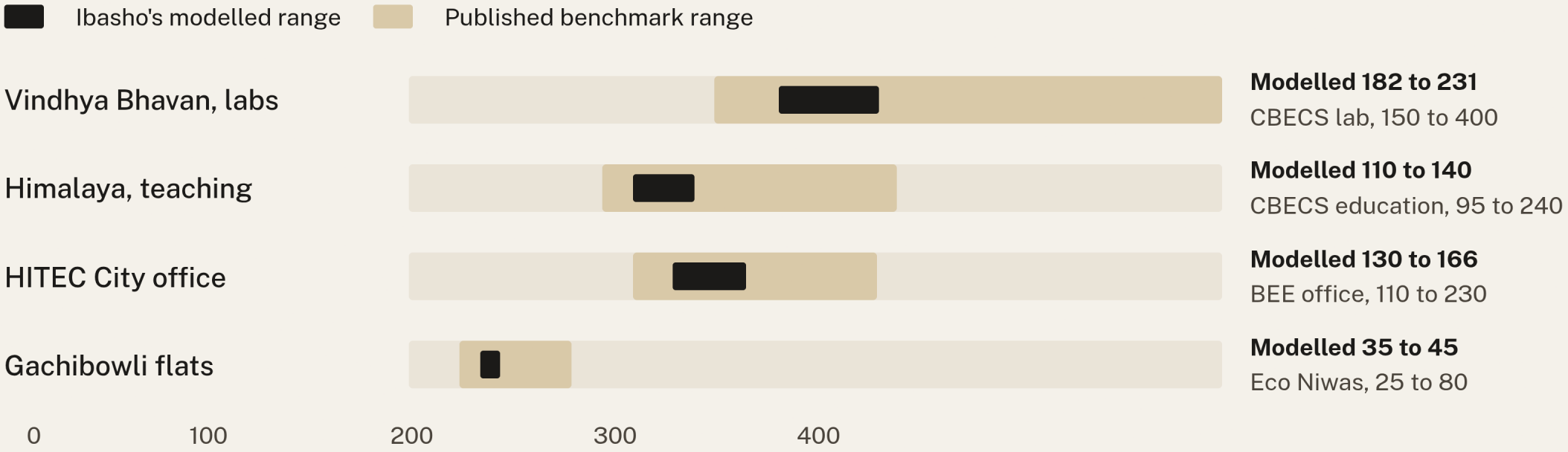
A narrator that explains
Plain words, every number sourced

Come back any time
Revisit, branch, update, see what changed

Works with AI agents
An MCP link for Claude Code and others

RESULTS: BUILDINGS

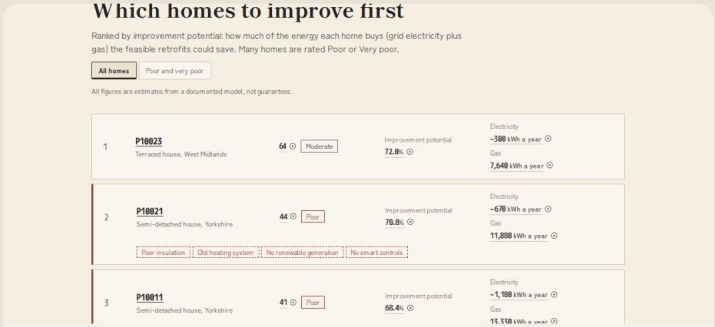
Modelled buildings land inside published benchmark ranges.



Energy use intensity, kWh per m^2 a year. Sun position matches the NREL reference to 0.00001°.

NEXT

From a working studio to real homes.



Real homes

Test the home model on real EPC records and meter data.



Full models in the studio

Switch on the full site, energy and code models in every view.



Grow the ladder

More cities, more building types, whole communities.

Ibasho

A better place to be, from a desk to a city.

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